

Chapter 12

Policy Effects and Cost Shocks in the AS/AD Model

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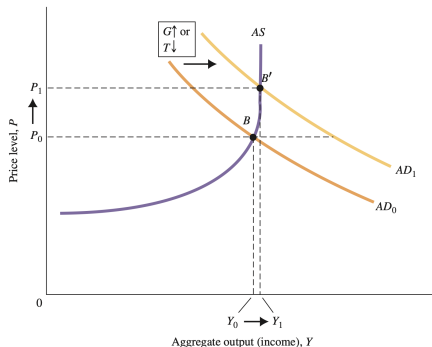
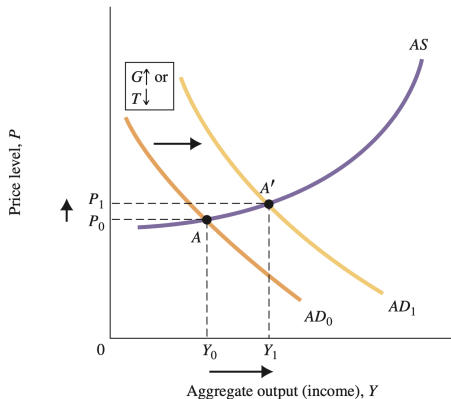
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Expansionary Fiscal Policy

- We turn now to look at government spending and taxes using the lenses of the AS/AD model.
- The effects of expansionary fiscal policy are different depending on the state of the economy on the AS curve.



The Fed's Responses to the Z Factors

- There are several features of the AS/AD model that we need to consider regarding the effectiveness of the Fed.
- Recall the Fed's rule

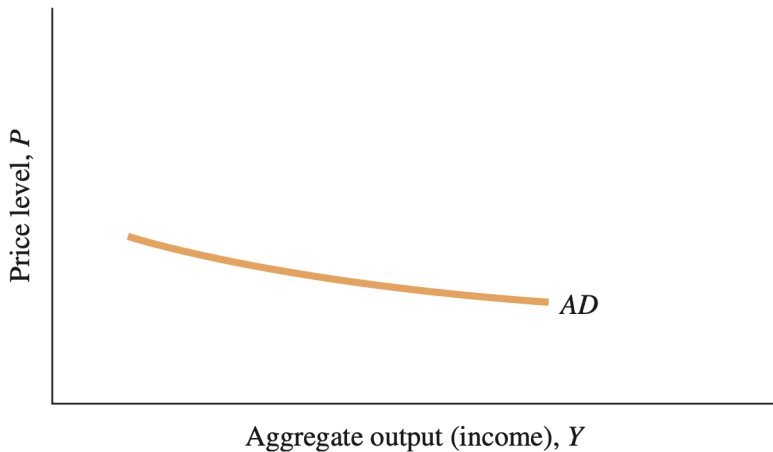
$$r = \alpha Y + \beta P + \gamma Z.$$

- The Fed is affected by exogenous variables, which are determined outside of the model, such as consumer sentiment and foreign economies
- An increase (decrease) in Z may prompt the Fed to increase the interest rate, thus shifting the AD curve to the left (right).

Shape of the AD Curve When the Fed Cares More About the Price Level than Output

- In the equation representing the Fed rule, we used a weight of α for output and a weight of β for the price level.
- The relative size of these two coefficients can be thought of as a measure of how much the Fed cares about output versus the price level.
- If $\alpha < \beta$, the AD curve is relatively flat, which means that the Fed is willing to accept large changes in Y to keep P stable.

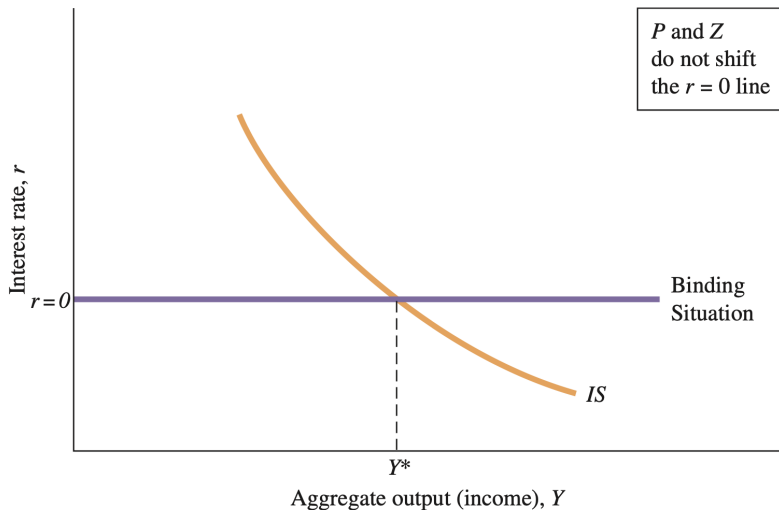
AD Curve when $\alpha < \beta$



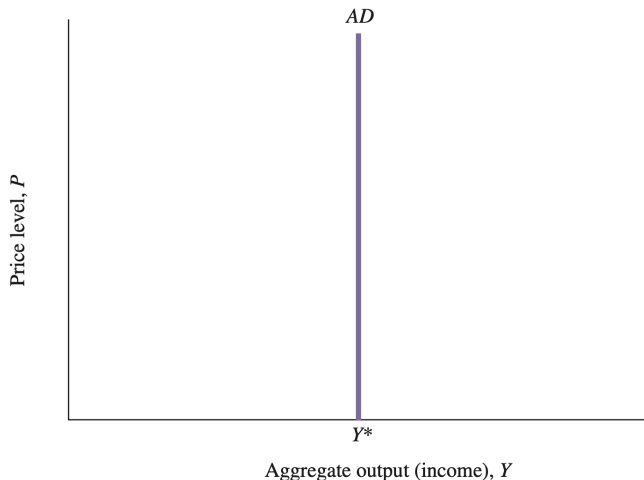
Zero Interest Rate Bound

- Note that nominal interest rate cannot go below zero.
- When the interest rate is zero and the Fed rule calls for a negative interest rate, we call this case a binding situation.
- In a binding situation, LM curve is horizontal and AD curve is vertical.
 - At zero lower bound, monetary policy is ineffective and fiscal policy is required.

LM Curve in a Binding Situation



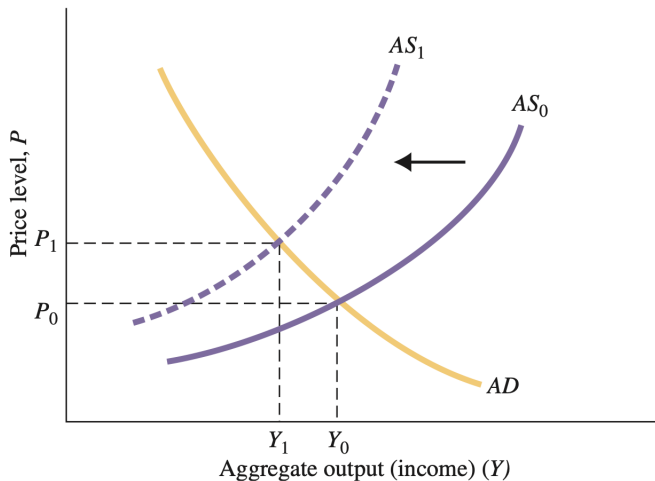
AD Curve in a Binding Situation



Cost (Supply-Side) Shocks

- A sudden and severe cold spell kills off a large fraction of the feeder-fish stock in the world.
- Heat waves increase the malfunctions of machines and destroy agricultural crops.
- Middle East war damages the oil supply regions.
- These examples of adverse cost shock are likely to increase all at once the costs of many firms in many different markets.
- The AS curve shifts to the left as firms who experience these new costs raise their prices to cover their new higher costs

Stagflation



Demand-Side Shocks

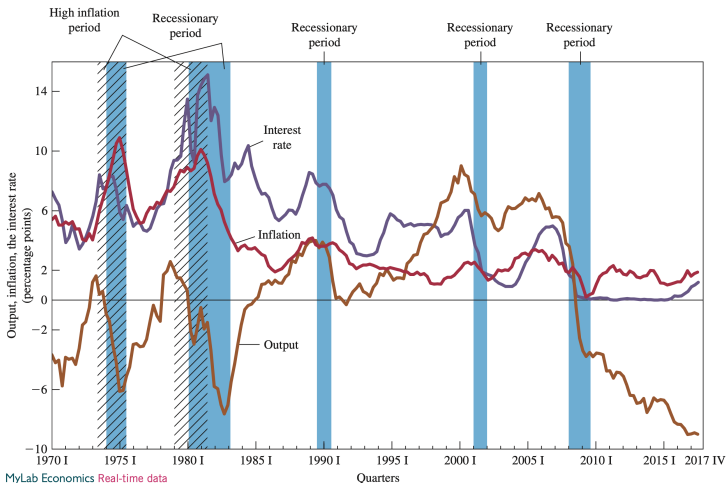
- Contrary to cost-push inflation, demand-pull inflation corresponds to higher output.
- *Animal spirits*, optimism or pessimism about the economy, can be thought of as demand-side shock.
- Any price increase that results from a demand-side shock is considered demand-pull inflation.

Expectations

- Animal spirits can be considered expectations of the future.
- Firms' expectations of future prices may affect their current price decisions.
 - An increase in future price expectations may thus shift the AS curve to the left and thus act like a cost shock.
- Because of the fact that expectations can affect the price level, expectations can lead to an inertia that makes it difficult to stop an inflationary spiral.
- Given the importance of expectations in inflation, the central banks of many countries survey consumers about their expectations.

Output, Inflation, and the Interest Rate

1970 I–2017 IV



Inflation Targeting

- A monetary authority chooses its interest rate values with the aim of keeping the inflation rate within some specified band over some specified horizon.
- For example, the target value might be 2 percent with a band of 1 to 3 percent.
 - Then the monetary authority would try to keep the actual inflation rate between 1 and 3 percent.
- There has been much debate about whether inflation targeting is a good idea.
 - It can lower fluctuations in inflation, but possibly at a cost of larger fluctuations in output.